Impedance matching and stubs

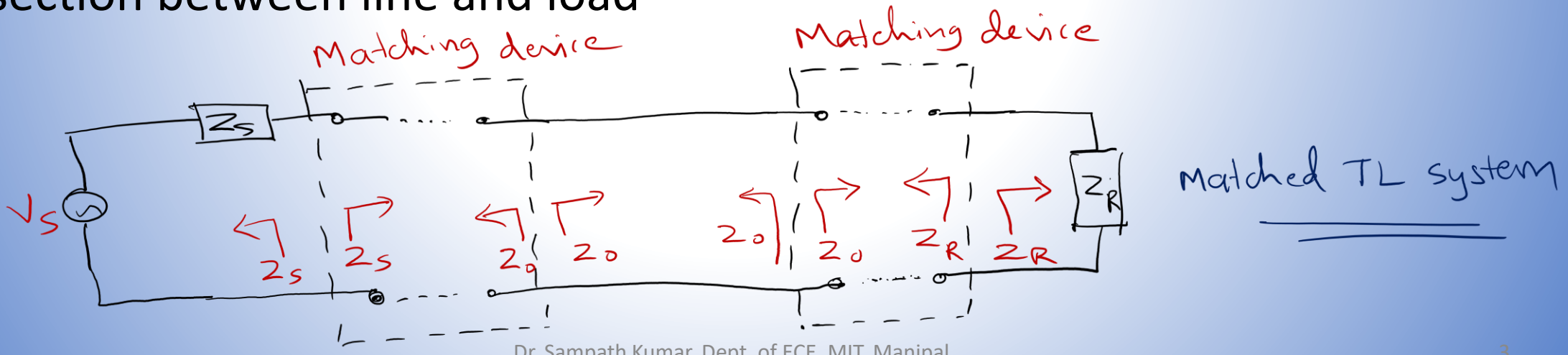
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Impedance matching:

- To get $Z_R = Z_s^*$ - maximum power transfer theorem
- Minimize reflection (Zero |K| or SWR unity)
- Optimize noise figure

Impedance matching:

- Standing waves lead to increased losses and frequently cause the transmitter to malfunction
- For greatest efficiency and delivered power, high frequency TL should be terminated with R_0
- However, loads such as antennas do not have resistance values equal to R_0
- It is necessary to introduce some form of impedance transforming section between line and load

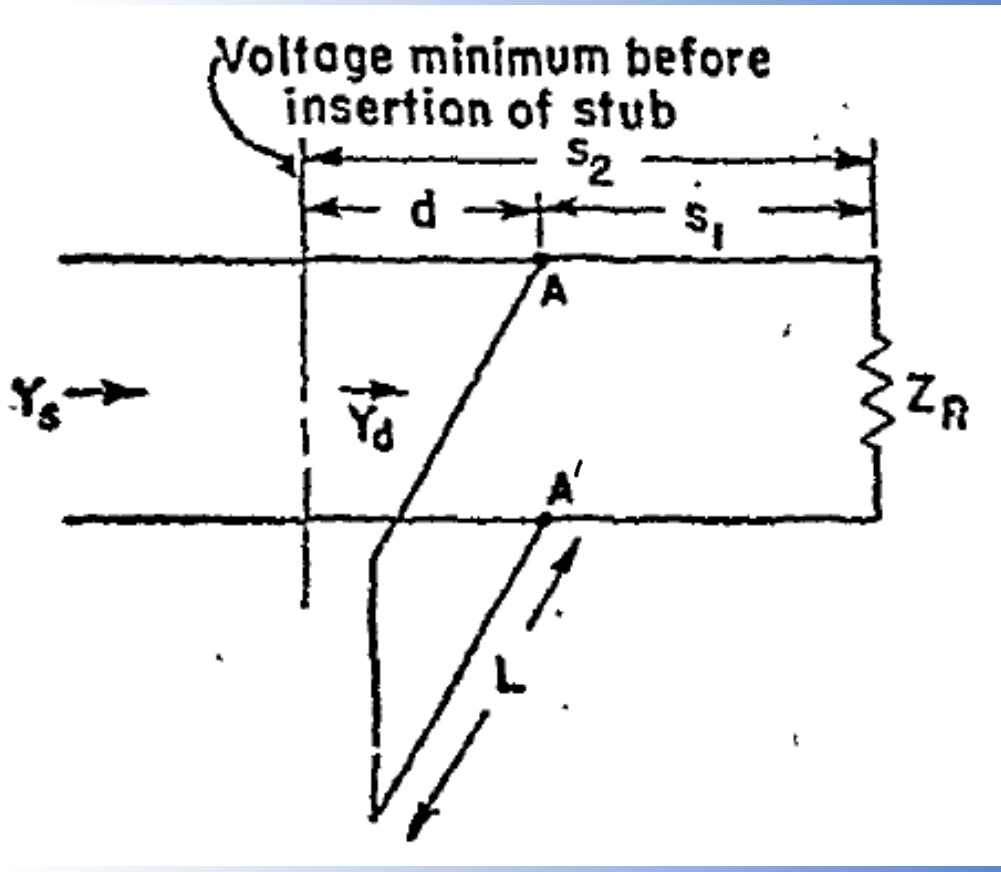


Single – stub impedance matching on a TL:

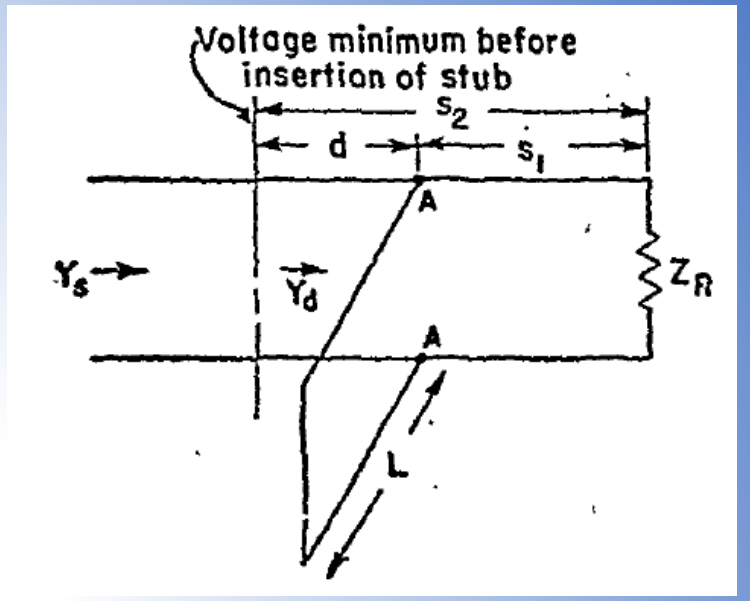
- The input conductance of a line is $1/SR_0$ at voltage maximum
- S/R_0 at voltage minimum
- At some intermediate point A, the real part of input admittance may have intermediate value $1/R_0$

OR input admittance at A has the value

$$Y_s = \frac{1}{R_0} \pm jB$$



After the point having conductance $= 1/R_0$ is located, a short stub line having input susceptance $\mp jB$ may be connected across TL

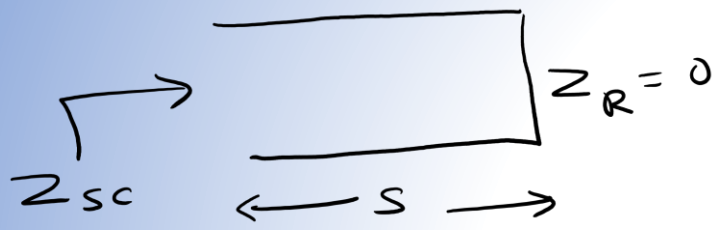


- The input admittance at point A is

$$Y_s = \frac{1}{R_0} \pm jB \mp jB = \frac{1}{R_0}$$

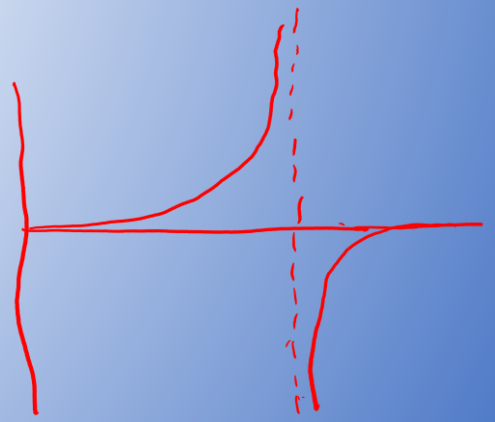
- The input impedance of TL at point A looking towards the load is $Z_s = R_0$
- The line from source to A is then terminated in R_0 and is a smooth line.
- From A to the load, reflection and standing wave occur.
- Since this distance can always be made less than a wavelength, the losses are not severe.

Short circuited stub:



$$Z_{sc} = jR_o \tan \beta s$$

- If the length of line is $0 < s < \lambda/4$ – inductive reactance (o to ∞)
- If $\lambda/4 < s < \lambda/2$ – capacitive reactance (o to -∞)
- Stubs are TLs whose length is very small i.e. less than $\lambda/2$
- It is used for obtaining the reactances from -∞ to ∞



Open circuited stub:

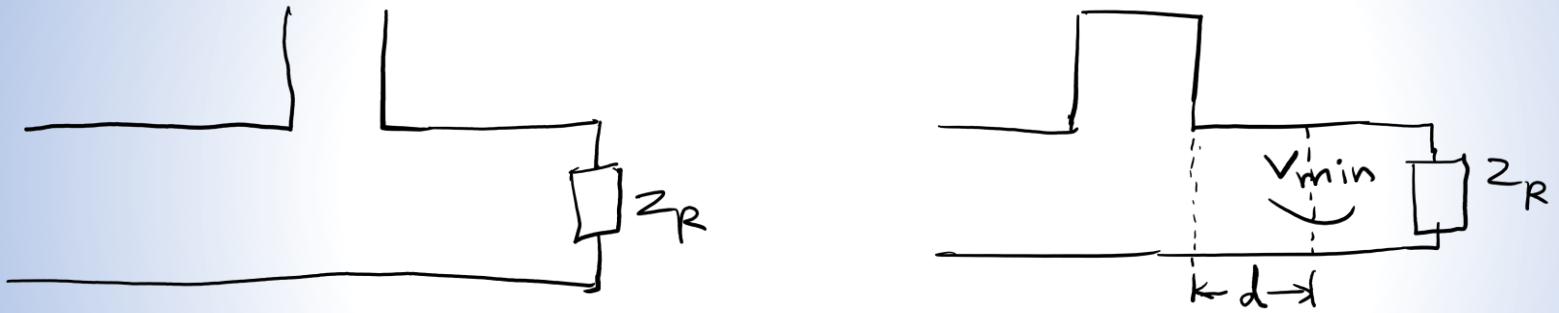
$$Z_{oc} = -jR_0C_0 + \beta s$$



Disadvantages:

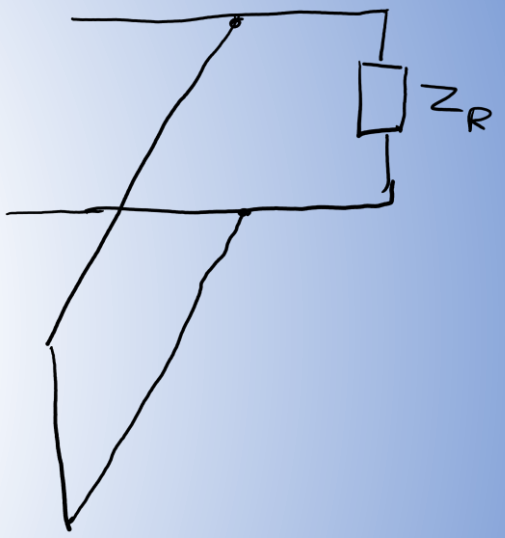
1. Construction is difficult – a good short circuit is easier to obtain than a good open circuit (At RF)
2. Power loss is present, since it acts an antenna and RF power radiates to the free space

Series stub:



Parallel stub – very commonly used

- Because of paralleling of elements, it is most convenient to work with admittance



$$Z_s = R_0\left(\frac{1 + |K|\underline{/\phi - 2\beta s}}{1 - |K|\underline{/\phi - 2\beta s}}\right).$$

$$Y_s = G_0\left(\frac{1 - |K|\underline{/\phi - 2\beta s}}{1 + |K|\underline{/\phi - 2\beta s}}\right) \quad \text{where } G_0 = \frac{1}{R_0}$$

$$\text{Normalized admittance } \frac{Y_s}{G_0} = \left(\frac{1 - |K|\underline{/\phi - 2\beta s}}{1 + |K|\underline{/\phi - 2\beta s}}\right)$$

Normalized admittance $\frac{Y_s}{G_0} = \left(\frac{1 - |K|/\phi - 2\beta s}{1 + |K|/\phi - 2\beta s}\right)$

Put $\theta = \phi - 2\beta s$

$$\frac{Y_s}{G_0} = \frac{1 - |k|\cos\theta - j|k|\sin\theta}{1 + |k|\cos\theta + j|k|\sin\theta} \times \frac{1 + |k|\cos\theta - j|k|\sin\theta}{1 + |k|\cos\theta - j|k|\sin\theta}$$

$$Y_s = G_0 \left[\frac{1 - |K|^2 - 2j|K|\sin(\phi - 2\beta s)}{1 + |K|^2 + 2|K|\cos(\phi - 2\beta s)}\right]$$

$$\frac{Y_s}{G_0} = \frac{G_s}{G_0} + \frac{B_s}{G_0} \quad \text{where}$$

$$\frac{G_s}{G_0} = \left[\frac{1 - |K|^2}{1 + |K|^2 + 2|K|\cos(\phi - 2\beta s)}\right] \rightarrow (1)$$

$$\frac{B_s}{G_0} = \left[\frac{-2|K|\sin(\phi - 2\beta s)}{1 + |K|^2 + 2|K|\cos(\phi - 2\beta s)}\right] \rightarrow (2)$$

$$\frac{G_s}{G_0} = \left[\frac{1 - |K|^2}{1 + |K|^2 + 2|K| \cos(\phi - 2\beta s)}\right]$$

$$\frac{B_s}{G_0} = \left[\frac{-2|K| \sin(\phi - 2\beta s)}{1 + |K|^2 + 2|K| \cos(\phi - 2\beta s)}\right]$$

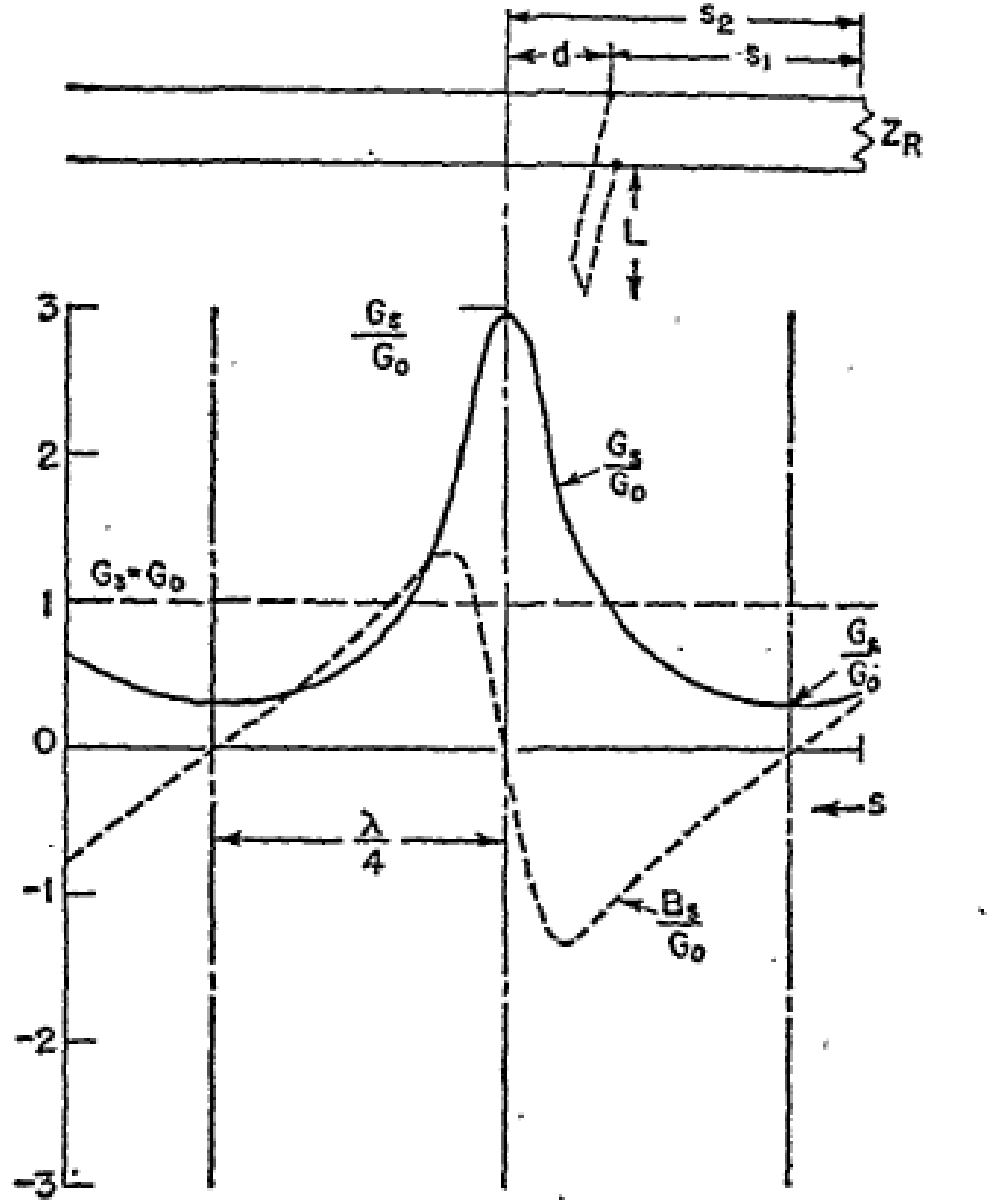


Fig. 7-18. Admittance conditions on a line indicating the proper location of the stub for $|K| = 0.5$.

- At $S = S_2$

$$\frac{G_s}{G_0} = \frac{1 - |K|^2}{1 + |K|^2 - 2|K|}$$

$$= \frac{1 + |K|}{1 - |K|} = S$$

i.e. when $\cos(\phi - 2\beta s_2) = -1$

or $\phi - 2\beta s_2 = \pm\pi$

$$s_2 = \frac{\phi + \pi}{2\beta}$$

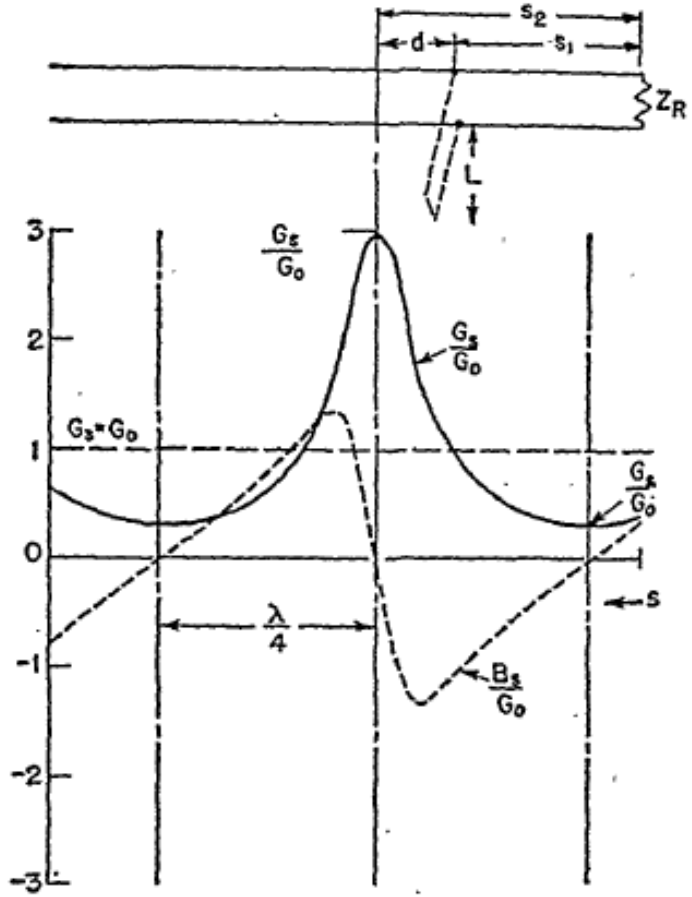


Fig. 7-18. Admittance conditions on a line indicating the proper location of the stub for $|K| = 0.5$.

At $S = S_2$ $\quad R_{min} = \frac{R_0}{S}$; $G_{max} = G_0 S$

$$\frac{G_s}{G_0} = \left[\frac{1 - |K|^2}{1 + |K|^2 + 2|K|\cos(\phi - 2\beta s)}\right]$$

- At $S = S_1$

$G_s = G_0$. This is the point at which the stub is to be connected.

The value of G_s/G_0 is unity there,

$$1 = \frac{1 - |K|^2}{1 + |K|^2 + 2|K| \cos(\phi - 2\beta s_1)}$$

$$\cos(\phi - 2\beta s_1) = -|K| \quad \text{----} \ (3)$$

$$\cos^{-1}(-|K|) = -\pi + \cos^{-1}|K|$$

$$s_1 = \frac{\phi + \pi - \cos^{-1}|K|}{2\beta}$$

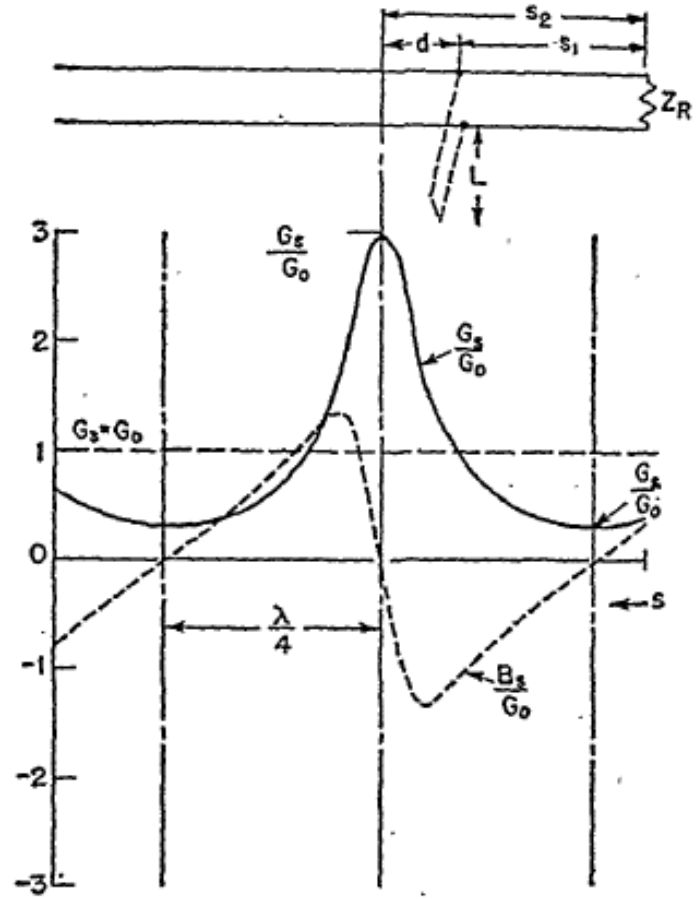


Fig. 7-18. Admittance conditions on a line indicating the proper location of the stub for $|K| = 0.5$.

Hence the distance d from the voltage minimum to the point of stub connection is $d = s_2 - s_1$

$$s_2 = \frac{\phi + \pi}{2\beta} \qquad s_1 = \frac{\phi + \pi - \cos^{-1}|K|}{2\beta}$$

$$d = \frac{\cos^{-1}|K|}{2\beta}$$

At $S = S_1$

$$\frac{B_s}{G_0} = \left[\frac{-2|K|\sin(\phi - 2\beta s_1)}{1 + |K|^2 + 2|K|\cos(\phi - 2\beta s_1)}\right]$$

From eq^n (3) $\cos(\phi - 2\beta s_1) = -|K|$

$$\sin(\phi - 2\beta s_1) = \sqrt{1 - \cos^2(\phi - 2\beta s_1)}$$

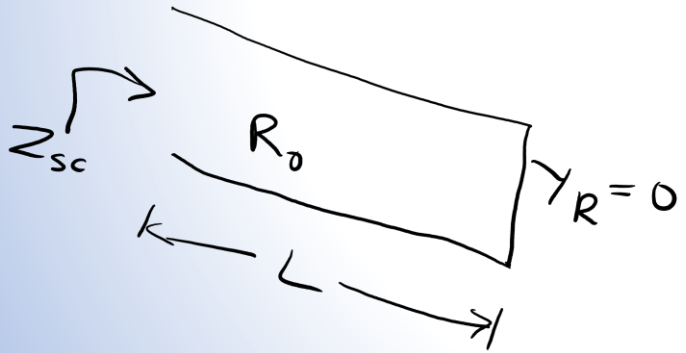
$$= \sqrt{1 - |k|^2}$$

$$\frac{B_s}{G_0} = \frac{-2|k|\sqrt{1-|k|^2}}{1+|k|^2-2|k|^2}$$

$$\frac{B_s}{G_0} = \frac{-2|k|\sqrt{1-|k|^2}}{1+|k|^2-2|k|^2}$$

$$= \frac{-2|k|\sqrt{1-|k|^2}}{1-|k|^2}$$

$$= \frac{-2|k|}{\sqrt{1-|k|^2}} \quad \text{—— (4)}$$



$$Z_{sc} = jR_0 \tan\beta s$$

$$Y_{sc} = -jG_0 \cot\beta s$$

$$= -jG_0 \cot\beta L$$

$$\frac{Y_{sc}}{G_0} = -j\cot\beta L$$

$$\frac{B_{sc}}{G_0} = -\cot\beta L \rightarrow (5)$$

- Compare (4) and (5)

$$\cot \beta L = \frac{2|k|}{\sqrt{1-|k|^2}}$$

$$\beta L = \tan^{-1}\left(\frac{\sqrt{1-|k|^2}}{2|k|}\right)$$

$$L = \frac{\lambda}{2\pi} \tan^{-1} \frac{\sqrt{1 - |K|^2}}{2|K|}$$

Substitute $|k| = \frac{S-1}{S+1}$ where S = SWR

$$L = \frac{\lambda}{2\pi} \tan^{-1} \frac{\sqrt{S}}{S - 1}$$

5 # Design a single stub matching network to match a load of $(200 - j150)\,\Omega$ to a line of $R_0 = 100\,\Omega$. The operating frequency is 30 MHz.

$$K = \frac{Z_R - R_0}{Z_R + R_0} = \frac{200 - j150 - 100}{200 - j150 + 100} = \frac{100 - j150}{300 - j150}$$

$$= \frac{180.27 \angle -56.3}{335.4 \angle -26.56}$$

$$K = 0.54 \angle -29.7^\circ$$

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{30 \times 10^6} = 10\,m$$

$$S_1 = \frac{\lambda}{4\pi}\left(\phi + \pi - \cos^{-1}|K|\right)$$

$$= \frac{10}{4\pi}\left(-29.7 + 180 - \cos^{-1}(0.54)\right) \times \frac{\pi}{180}$$

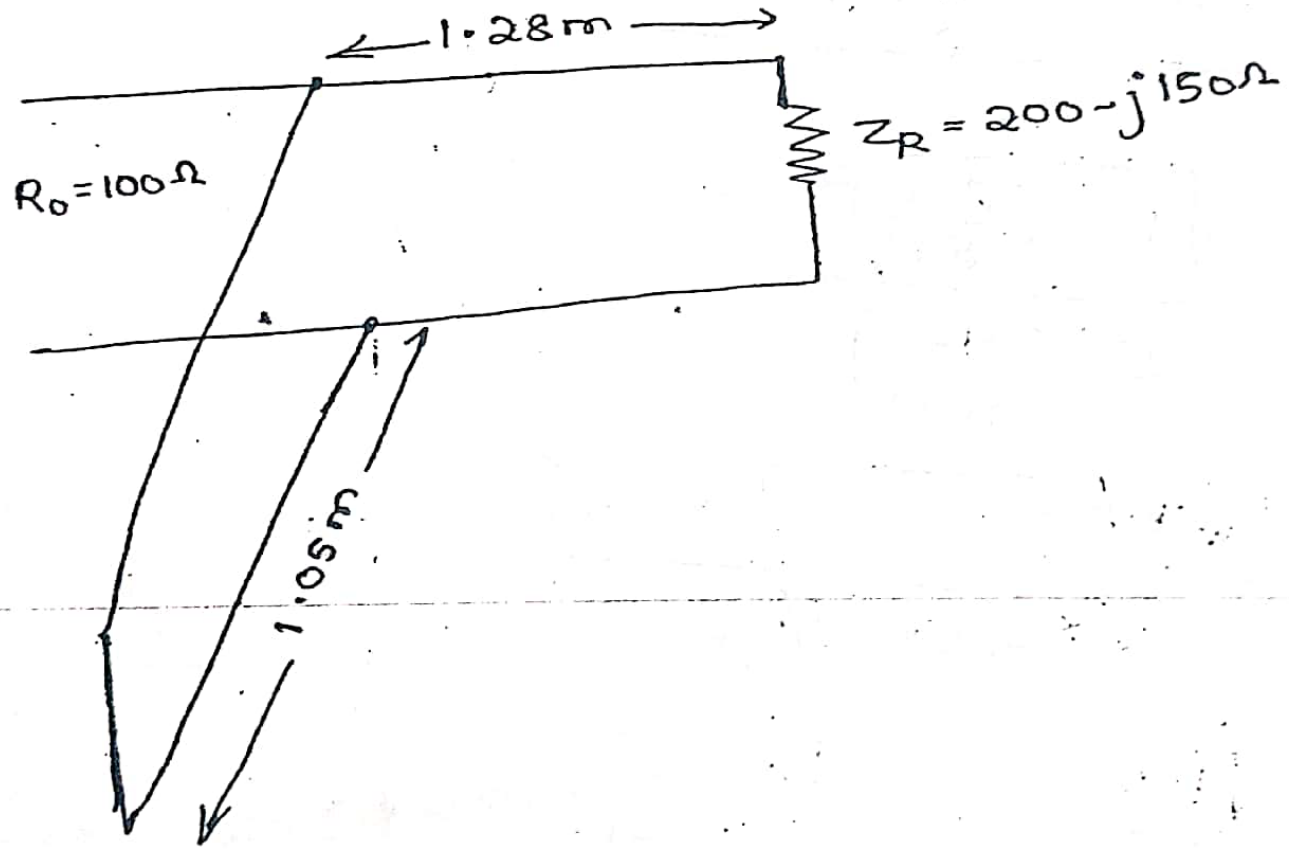
$$= \frac{10}{4 \times 180}\left(-29.7^\circ + 180^\circ - 57.52^\circ\right)$$

$$\boxed{S_1 = 1.28\,m}$$

$$L = \frac{\lambda}{2\pi} \tan^{-1} \frac{\sqrt{1-|K|^2}}{2|K|}$$

$$= \frac{10}{2\times 180} \tan^{-1} \frac{\sqrt{1-0.54^2}}{2\times 0.54}$$

$$\boxed{L = 1.05\ m}$$



Bring 4 Smith charts and Compass box for next class


Thank
you